

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

FACULTY: DR.D.SUDHAGAR

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks: 50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

The large difference between the **training Mean Squared Error (MSE = 0.85)** and the **testing Mean Squared Error (MSE = 5.20)** indicates that the neural network performs very well on the training data but poorly on unseen test data. This is a sign of **overfitting**, where the model memorizes the training data instead of learning general patterns.

Reasons for the Difference

1. **Overfitting**
 - The model has learned the training data too closely, including noise and irrelevant details.
 - It fails to generalize to new student data.
2. **Insufficient Training Data**
 - A small dataset may not represent all types of student performance, leading to poor predictions on new data.
3. **Model Complexity**
 - Too many hidden layers or neurons increase the model's capacity, causing it to fit the training data excessively.
4. **Noisy or Biased Data**
 - Errors, missing values, or outliers in the training dataset can reduce prediction accuracy on the test set.
5. **Improper Data Splitting**
 - If the training and testing datasets have different characteristics, the model cannot perform well on the test data.
6. **Lack of Regularization**

- Without techniques to prevent overfitting, the model becomes too specialized for the training dataset.

Corrective Measures

1. **Use More Training Data**
 - Collect additional student records or use data augmentation techniques to improve learning.
2. **Apply Regularization**
 - Use **L1/L2 regularization** to reduce overfitting by penalizing large weights.
3. **Use Dropout**
 - Randomly deactivate neurons during training to improve the model's generalization ability.
4. **Simplify the Model**
 - Reduce the number of hidden layers or neurons to avoid unnecessary complexity.
5. **Early Stopping**
 - Stop training when the validation error starts increasing to prevent overfitting.
6. **Cross-Validation**
 - Use k-fold cross-validation to evaluate model performance on different subsets of data.
7. **Data Preprocessing**
 - Remove outliers, handle missing values, and normalize input features to improve model performance.

Conclusion

Since the **training MSE (0.85)** is much lower than the **testing MSE (5.20)**, the neural network is **overfitting** the training data. Applying regularization, dropout, early stopping, cross-validation, increasing the dataset size, and simplifying the model can reduce the error gap and improve prediction accuracy on unseen student data.

Code of Conduct:

I M.Pranai(192525068) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.Pranai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided